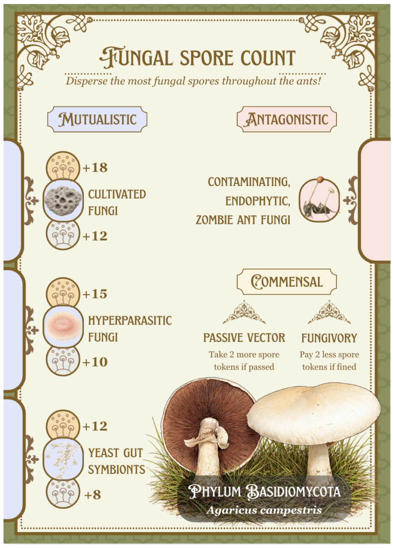
Serious Game for Fungal Education:
🍄 Spore Smugglers: Inspection of the Queen Ant 🐜

Elren Yeongin Chae



This is a serious board game titled *Spore Smugglers: Inspection of the Queen Ant*, a bluffing and negotiation game inspired by the famous board game *Sheriff of Nottingham*. Players take on the role of fungi merchants attempting to spread different fungal species past a rotating "Queen Ant" inspector by concealing cards, making strategic declarations, and negotiating outcomes. The game includes a deck of fungi cards representing real species, player boards, spore tokens, and bags used to hide selected cards each round.

The game is designed to educate players about fungal biology and ecology, with a focus on the diverse relationships between fungi and ants. Scientific names are intentionally included to familiarize players with biological nomenclature through repeated use during gameplay. Each card features realistic illustrations that correspond to the correct taxonomy or fungal group, reinforcing visual recognition alongside scientific classification. The game also highlights the diversity of fungi beyond the typical mushroom form, introducing players to a wide range of morphologies such as molds, yeasts, and parasitic fungi.

Through gameplay, players encounter mutualistic fungi that support ant colonies, antagonistic fungi that threaten them, and context-dependent fungi that interact in more indirect ways. These key ecological concepts of symbiosis, parasitism, commensalism, and spore dispersal are embedded directly into the game mechanics. By closely modeling real-world interactions, this design gamifies ecological relationships into memorable and effective learning, with the ultimate goal reflecting the fungal biology of successfully dispersing as many spores as possible. By shedding light on a specific topic as the fungi-ant relationship, this game also aims to spark curiosity within game players on how diversely interesting fungi biology and ecology can be.

Game introduction

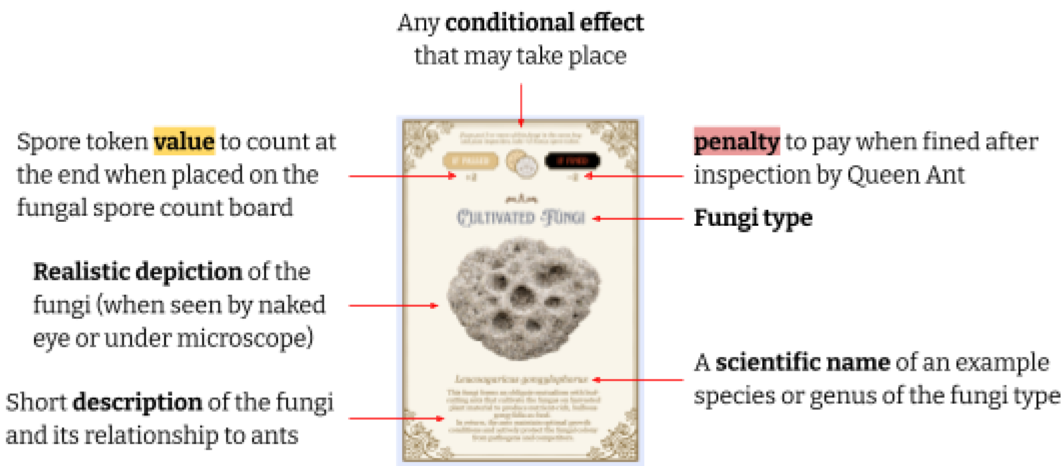
A 3–5 player bluffing game where players act as **fungi merchants** trying to spread as many fungi past the **Queen Ant (inspector)** and proliferate in the ant queendom and beyond. The game is grounded in real ecological relationships between fungi and ants across:

- Mutualism (beneficial to both)
- Antagonism (helpful in spreading fungal spores but harmful to ants)
- Contextual/commensal (neutral or indirect)

Players bluff, negotiate, and strategically combine fungi to maximize **spore points**. Whoever has the most spore points at the end wins.

Materials & Setup

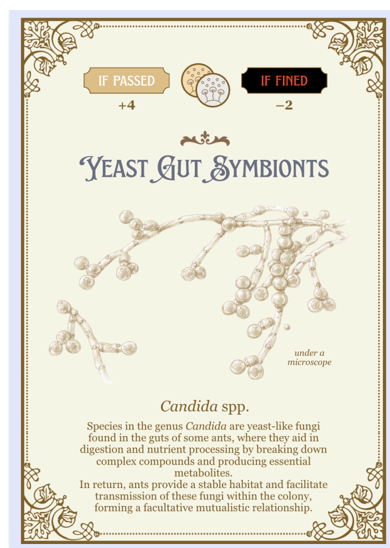
- 204 fungi cards of 8 types, each with:



Legal cards (mutualistic): safe to declare as import

Total 50 cards	Total 40 cards	Total 40 cards

The cultivated fungi has the highest number of cards amongst mutualistic fungi to mimic how it is one of the most well-established mutualistic fungi-ant relationships.



Contraband (antagonistic): must be smuggled

Total 24 cards

Total 20 cards

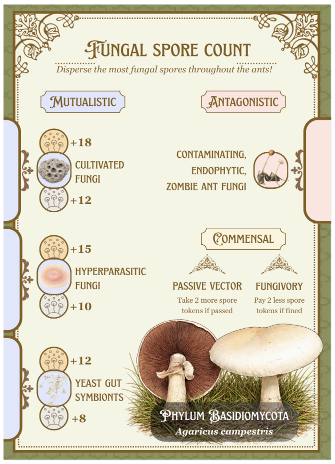
Total 16 cards

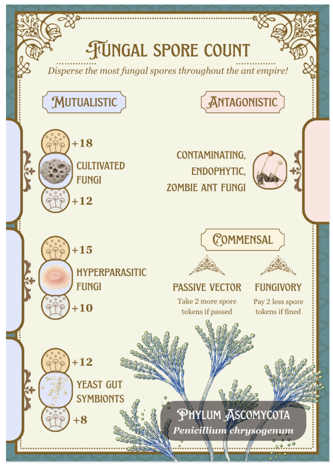
Special (contextual / commensal): modify outcomes when included in bag

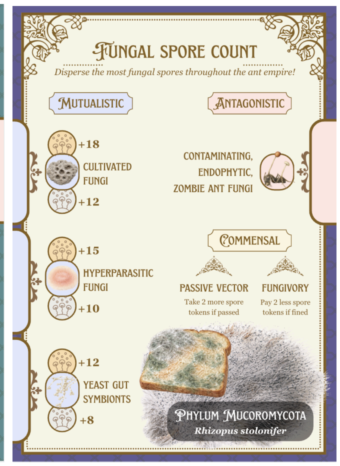
Total 7 cards

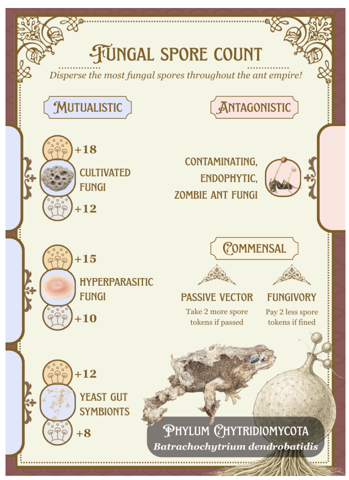
Total 7 cards

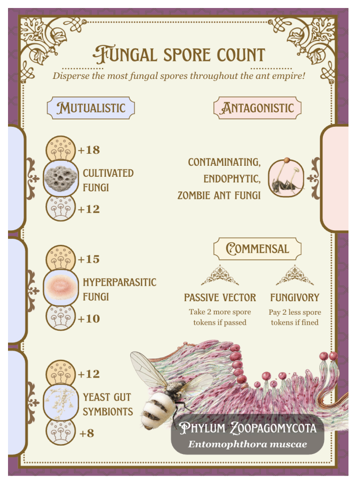
- 5 player boards (each named with phylum)
  - Similar to how the board game *Sheriff in Nottingham* has non-narratively vital characters on its player boards, this game uses the boards as an opportunity to showcase some of the well-researched phyla of the fungi kingdom.
  - The fungal spore count boards indicate where the players should place their passed fungi cards, special rules for contextual / commensal cards, and a depiction of an example fungal species of a phylum.











- 5 merchant bags for each player to conceal their cards when handing over to the Queen Ant for possible inspection
- 1 Queen Ant marker to indicate who the round's Queen Ant is
- Spore tokens (~100) that work as coins

Rules

<u>Round Structure</u>

1. Set-up
    - Each player starts with 6 cards, 50 spores, 1 bag, and 1 board
    - Shuffle deck → deal cards → choose player to start as Queen Ant
2. Draw & Bag
    - Discard up to 5 cards, draw back to 6
    - Place 1-5 cards in bag
        - It may be any combination of fungi cards, but be ready to lie.
3. Declare
    - State: "I have ___ cards [number of cards] of ___ fungi [one type of legal fungi]"
    - Must declare only **one type**, and one legal fungi
    - This statement may be truthful or be a lie (you have other types of mutualistic fungi, antagonistic, or commensal cards mixed in)
    - e.g. "I have 4 cards of cultivated fungi"
4. Negotiate
    - Offer spore tokens, card exchanges, or promises (that may never be kept) to the Queen Ant to avoid inspection
    - Strike a deal with the Queen Ant that if they let you pass now, you will also let them pass in the next round when you play Queen Ant
    - Bluff that they'll regret it and have to pay the fines if they inspect you
5. Inspection (Queen Ant decides)
    - Either inspect bag (reveal cards inside) or allow bag to pass for each player bag

<u>If inspected by Queen Ant</u>

- **If truthful** → Queen Ant pays penalty (sum of penalties on each card), player keeps all cards and places them on their fungi spore count board
    - e.g. Fungi merchant declares that they have 4 cards of cultivated fungi
        - → Queen Ant suspects they are lying and inspects
        - → revealed in bag that there are indeed 4 cultivated fungi cards
        - → Queen Ant pays 4 x 2 = 8 spore tokens to the fungi merchant;
        - the fungal merchant can place the 4 cultivated fungi cards on their fungal spore count board
- **If lying** → player pays penalty (sum of penalties on contraband cards), confiscate all undeclared cards, declared ones pass and are placed on fungi spore count board
    - e.g. Fungi merchant declares that they have 4 cards of cultivated fungi
        - → Queen Ant suspects they are lying and inspects
        - → revealed in bag that there are 2 cultivated fungi, 1 hyperparasitic fungi, and 1 contaminating fungi
        - → fungi merchant pays 2 (for undeclared card, the hyperparasitic fungi) + 4 (undeclared contraband card, the contaminating fungi) = 6 spore tokens to the Queen Ant;
        - these 2 undeclared fungi are confiscated (returned to deck) and the fungal merchant can place 2 cultivated fungi that was declared on their fungal spore count board

- **Special Effects**
  *modify outcomes but do not override core rules (confiscation, inspection, penalties)
    - Your inspected bag had a **contaminating fungi** (contraband / antagonistic):
        - → **Destroy 1 cultivated fungi** in the same bag or on your board, if present (additional penalty for fungi merchant)
        - *Metaphor on ecological relationship:*
            - Contaminating fungi are specialized parasites that invade and degrade ant fungal gardens. It may attack existing hyphae of fungi such as *Leucoagaricus gongylophorus*. The penalty of getting caught smuggling contaminating fungi translates into your overall fungi spore dispersal being cancelled out because the contaminating fungi attacked your cultivated fungi.
    - Your inspected bag had a **hyperparasitic fungi** (legal / mutualistic):
        - → **Ignore penalty of 1 antagonistic fungus** in the same bag, if present (advantage for fungi merchant)
        - *Metaphor on ecological relationship:*
            - Hyperparasitic fungi are generalist mycoparasites that suppress other fungi, reducing their harmful impact. Although you got caught by the Queen Ant, this hyperparasitic fungi helped suppress one of your antagonistic fungi in the bag if there was one.
    - Your inspected bag had a **fungivory** (special / commensal):
        - → **Reduce total penalty by 2 spore tokens** (advantage for fungi merchant)
        - *Metaphor on ecological relationship:*
            - This special card is beneficial to ants who may consume mushrooms as food, while neutral or even antagonistic to fungi who are preyed on. Having this in your bag when inspected by the Queen Ant means you were trying to do good for the ants; hence, getting caught for this card translates to you being less heavily penalized.

<u>If NOT inspected (passed) by Queen Ant</u>

- Player keeps all cards, including contraband cards
    - place contraband face-down on the fungal spore count board
- **Special Effects**
  *modify outcomes but do not override core rules (confiscation, inspection, penalties)
    - Your passed bag had a **passive vector** (special / commensal):
        - → **Gain +2 bonus spore token**
        - *Metaphor on ecological relationship:*
            - Fungi produce abundant spores that may disperse via insects like ants as passive vectors. When you've successfully imported a passive vector card, it means you've managed to spread more spores.
    - Your passed bag had **3 or more cultivated fungi** (legal / mutualistic):
        - → **Gain +1 bonus spore token**
        - *Metaphor on ecological relationship:*

- These fungi are cultivated in large quantities by ants for food. Importing 3 or more of these fungi cards into the ant queendom is like a farming technique beneficial to both fungi and ants.

End of Round

- Players refill from the card deck to have 6 cards in their hands
- The player on the left of the current Queen Ant plays the next Queen Ant

Final Scoring & Winning

- The game ends after each player has been Queen Ant an equal number of times
  - 3 players: 3 Queen Ant turns each
  - 4-5 players: 2 Queen Ant turns each
- Each player should count:
  - Total amount of spore tokens
  - Total value written on each card on the fungal spore count board
- Also count in the bonuses:
  - Most of each legal card type → gain spore token indicated in Gold bonus
  - Second most → gain spore token indicated in Silver bonus
- Congratulations! The player with the highest total **spore points** wins for dispersing the most fungal spores into the ant queendom and beyond.

Resources

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